

TRAVANCORE RESEARCH & INVESTMENTS LTD.

Invest · Trade · Grow | Trade with Insight

COMPREHENSIVE MODULE-BY-MODULE REQUIREMENT & SPECIFICATION DOCUMENT FOR FULL-STACK DEVELOPMENT

Target Technology Stack:	PostgreSQL (Database), Sequelize (ORM), Node.js / Express (Backend API), React.js (Mobile-Responsive SPA Frontend)
Design Concept:	Zerodha-inspired Demat / Trading Application Interface with Deep Navy Blue, White, and Gold Visual Identity
Source Document:	FOR EXECUTING_4.pdf (25 Comprehensive Wireframes & Functional Guidelines)
Compliance Policy:	Strict exclusion of mock regulatory registration numbers (SEBI/NSE/BSE/MCX/CDSL). Protected by secure authentication.

1. Executive Summary & Core Platform Vision

Travancore Research & Investments Ltd. is a state-of-the-art web and mobile-responsive financial trading platform. The objective of this application is to deliver a familiar, ultra-intuitive, and high-trust Demat user experience (similar to leading trading apps like Zerodha). The system seamlessly combines manual watchlist trading, automated Algo Trading strategies, detailed P&L; analytics, multi-tiered account onboarding, referral earnings payouts, research advisory, and secure funds management.

Key Platform Capabilities:

- Multi-Modal Quick Authentication:** 6-digit transaction PIN, Fingerprint, Face ID, Pattern Lock, and traditional Password authentication.
- 7-Step Seamless Onboarding Wizard:** Comprehensive KYC, Bank details verification, and Trading package customization.
- Automated Algo Trading Engine:** One-click Master Algo ON/OFF toggle, strategy selection, live execution monitoring, and win/loss analytics.
- Multi-Market Support:** Indian Equity & F&O; Markets, Commodity Markets, and Crypto Asset Markets with real-time tickers.
- Referral Profit-Sharing System:** Automated 5% profit share on referee trades for up to 2 years with instant payout to bank or wallet.
- Institutional Financial Reporting:** Exportable P&L; reports, daily performance trend charts, and tax summary statements.

2. Module 1: Brand Design System & UI/UX Requirements

UI Element	Specification & Design Standard
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Color Palette	Primary Navy: #003366 Royal Blue: #0047AB Gold Accent: #D4AF37 Success Green: #10B981 Alert Red: #EF4444 Background: #FFFFFF & #F4F6F9
Header Branding	Travancore emblem featuring Padmanabhaswamy Temple logo, growth arrow chart icon, and tagline 'Trade with Insight'.
Responsive Layout	Mobile (<768px): Fixed top header, single column layout, sticky bottom navigation bar (Home, Markets, Algo Trade, Funds, More). Desktop (>1024px): Multi-column dashboard layout with sidebar navigation.
Typography & Feel	Clean sans-serif typography (Inter / Roboto / System), crisp card shadows, smooth tab transitions, and high-trust financial styling.
Regulatory Disclaimer	Strict prohibition of fake SEBI/NSE/BSE registration numbers. Display neutral security badge: 'Your account is protected with secure authentication.'

3. Module 2: Authentication & Security Management (Pages 3–6, 24)

2.1 Functional Requirements:

1. Welcome Back & Account Switcher Screen:
 - Display active profile avatar, Name (e.g., 'Sachin Tendulkar'), and Account ID ('458921').
 - Provide a prominent 'Change Account' button to switch between multiple registered trading profiles.
 - Quick access links at footer: *Privacy Policy* | *Terms & Conditions* | *Help & Support*.
2. Primary Quick Authentication Options:
 - 6-Digit Transaction PIN: Fast keypad entry with hashed verification.
 - Biometric Authentication: Integrated WebAuthn API for Fingerprint and Face ID authentication.
 - Pattern Lock: 3x3 grid gesture pattern verification.
 - Password Fallback: Standard Username + Password login form with 'Remember Me' and 'Forgot Password' link.
3. Security Settings & Audit Log (Page 24):
 - Account Status Banner: Live protection badge showing last successful login timestamp (e.g., 'Last login: 12 Sep 2026, 09:14 AM').
 - Credential Management: Change Password modal, Change 6-Digit PIN modal, Set/Update Pattern Lock.
 - Biometric Toggles: Master toggle switches to enable/disable Fingerprint Login and Face ID Login.
 - Login History & Audit: Chronological log table recording Login Date, Time, IP Address, Device Name, and Auth Method.
 - Trusted Devices & Revocation: View active sessions across mobile/desktop browsers with one-click 'Logout from All Devices' action.

4. Module 3: 7-Step Onboarding & Account Creation Wizard (Pages 7–14)

The onboarding module guides new users through a 7-step account opening process. Includes a step progress tracker bar, input validation, document upload simulation, and collapsible review sections.

Step #	Step Name	Required Fields & Functional Specification
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Step 1	Personal Details	• Full Name (as per official ID card) • Date of Birth (DD/MM/YYYY date picker) • Gender selection (Male, Female, Other)
Step 2	Contact Details	• Mobile Number (+91) with instant SMS OTP verification trigger • Email Address with verification link / code notification
Step 3	Identity & KYC	• PAN Card Number (Formatted string validation) • Aadhaar / Official ID Number (Optional / Formatted) • File Upload widget: Upload PAN/Aadhaar/Passport (JPG, PNG, PDF max 5MB) with preview
Step 4	Address Details	• Residential Street Address (House No, Street, Area) • City / District • State dropdown selector • 6-Digit PIN Code • Country (Default: India)
Step 5	Investor Profile	• Occupation dropdown (Salaried, Self-Employed, Business, Student, Retired) • Annual Income Range dropdown (< █1L, █1L-5L, █5L-10L, █10L-25L, > █25L) • Investment Experience level (Beginner, Intermediate, Expert) • Risk Profile (Capital Protect / Low, Moderate, High) • Investment Objective (Wealth Creation, Monthly Income, Capital Growth)
Step 6	Bank Details	• Account Holder Name (Must match Step 1 Name) • Bank Account Number & Confirm Account Number • IFSC Code entry with auto lookup ('Search IFSC') -> Auto-populates Bank Name and Branch
Step 7	Trading Setup & Review	1. Investment Package Selection: • <i>Capital Protect</i>: 20% - 30% Return/Month (Steady, Safe, Smart) • <i>Balanced</i>: 30% - 60% Return/Month (Stability, Opportunity, Growth) • <i>Aggressive</i>: 50% - 70% Return/Month (High Volatility, High Potential) 2. Markets & Instruments Selection: • <i>Indian Markets</i>: NIFTY, SENSEX, BANKNIFTY, FINNIFTY, STOCKS, OPTIONS (09:00 AM - 04:00 PM) • <i>Commodity Markets</i>: GOLD, SILVER, CRUDE OIL, NATURAL GAS, COPPER (06:00 PM - 12:00 AM) • <i>Crypto Markets</i>: BTC, ETH, SOL, USDT, BNB (24x7 / 09:00 AM - 12:00 PM) 3. Accordion Summary & Declarations: Edit buttons for all steps, consent checkboxes, and 'Create Account' submit button.

5. Module 4: Home Dashboard & Algo Trading Engine (Pages 15, 19)

4.1 Home Dashboard Layout & Components:

- **Header Strip:** Travancore logo, notification bell with unread badge counter, active user dropdown profile ('Sachin Tendulkar', ID: 458921).
- **Live Index Ticker Bar:** Real-time streaming bar showing NIFTY 50 (25,017.35 | +0.65%), SENSEX (81,697.76 | +0.64%), BANK NIFTY (51,328.45 | +0.61%), live date/time indicator, and 'Market Open' green status badge.
- **Master Algo Trading Control Card:** Large toggle switch (ON / OFF). Displays active status: 'Algo Trading is Active' or 'Market Closed / Session Completed'.
- **Portfolio Overview Cards:** Total Investment (█ 5,00,000), Current Portfolio Value (█ 5,38,750), Overall P&L; (+ █ 38,750 | +7.75%) accompanied by an interactive green performance curve graph.
- **Your Trading Package Banner:** Displays active package (e.g. Balanced 30%-60%) with a direct 'Change Package' button.
- **Markets & Instruments Card:** Active market tags (Indian Markets, Commodity, Crypto) with 'Manage Selection' modal button.
- **Quick Actions Grid:** 4 action buttons: + Add Funds, █ Withdraw, Account Details, and Reports.

4.2 Live Algo Trading Monitor (Page 19):

- **Live Execution State:** Toggle switch with live green indicator ('Live Execution') or red indicator ('Market Closed').
- **Active Positions List Table:** Symbol, Qty, Avg Buy Price, LTP, Live P&L; (■), and Status ('Open'). Example items: NIFTY 12 SEP 25000 CE (+■12,900), BANKNIFTY 12 SEP 51300 PE (-■4,150), RELIANCE (+■5,875), GOLD OCT FUT (+■1,620), BTC/USDT (+■620).
- **Session Summary Card:** Today's Live P&L; (+ ■ 16,865 | +3.37%), Total Trades Executed (8), Open Positions (5), Winning Trades (6 | 75% Win Rate).

6. Module 5: Markets Watchlist & Trading Instruments (Page 16)

5.1 Market Watchlist Specifications:

- **Categorized Filter Tabs:** All, Indian Markets, Commodities, Crypto, Top Gainers, Top Losers.
- **Search & Filter Bar:** Instant keyword search input ('Search for a stock, index, commodity or crypto...') with a **Filters** button for asset class, exchange, and volume.
- **Market Instrument Rows:**
 1. **Indian Markets:** NIFTY 50, SENSEX, BANK NIFTY, RELIANCE (■2,914.75 | +1.58%), TCS (■3,789.10 | +0.54%). Hours: 09:00 AM - 04:00 PM, Mon-Fri.
 2. **Commodities:** GOLD (■72,358 | +0.39%), SILVER (■83,420 | +0.74%), CRUDE OIL (■6,847 | -0.26%), NATURAL GAS (■245.30 | +1.16%). Hours: 06:00 PM - 12:00 AM, Mon-Fri.
 3. **Crypto Markets:** BITCOIN (\$58,421.35 | +2.18%), ETHEREUM (\$2,432.10 | +1.61%), SOLANA (\$142.35 | -1.25%). Hours: 24x7 / All 7 Days.
- **Action Context Menu (3 dots):** Buy, Sell, View Chart, Add to Favorites, Price Alert trigger.

7. Module 6: P&L; Analytics & Detailed Algo Reports (Pages 17, 20)

6.1 P&L; Dashboard Requirements (Page 17):

- **Time Filters:** Today, Daily, Weekly, Monthly, Custom Date Range picker.
- **P&L; Metric Cards:** Total Investment (■ 5,00,000), Current Value (■ 5,38,750), Overall P&L; (+ ■ 38,750 | +7.75%), Realized P&L; (+ ■ 27,850), Unrealized P&L; (+ ■ 10,900), Total Trades (12), Win Rate (66.7%).
- **P&L; Trend Graph:** Area chart mapping daily timeline (09:15 to 15:30) with cumulative profit trajectory.
- **Data Tables:** Positions (3), Closed Positions, Day-Wise P&L;, Instrument-Wise P&L;.
- **Report Export Engine:** Button dropdown 'Export' to generate PDF P&L; Statements, Excel Trade History, and Tax Reports.

6.2 Detailed Algo Performance Report (Page 20):

Report Section	Metrics & Data Visualization Specifications
Strategy Header	Selected Strategy: Balanced (30%-60% Target) Markets: Nifty, Bank Nifty, Stocks, Options Status: Active
Performance KPIs	Algo Investment: █ 5,00,000 Total Trades: 28 Winning Trades: 21 Losing Trades: 7 Win Rate: 75.0% Gross P&L: █ 52,340 Net P&L: █ 48,625 (after charges) ROI: +9.73%
Daily Chart & Highlights	Bar & Line Combo Chart (Profit vs Loss bars + Cumulative P&L; line). Best Day: + █ 14,320 (5 Sep 2026) Worst Day: - █ 5,860 (3 Sep 2026)
Trade Outcome Donut	Donut chart illustrating Win Ratio: 21 Wins (75.0% Green) vs 7 Losses (25.0% Red).
Breakdown Data Tables	Tabs: <i>Trade Details</i> <i>Day Wise P&L</i> ; <i>Month Wise P&L</i> ; <i>Instrument Wise</i> <i>Strategy Wise</i> . Columns: Date, Total Trades, Winning, Losing, Win Rate %, Gross P&L;, Net P&L;.
Actions & Scheduling	Download Report (PDF/Excel), Share Report (Email/WhatsApp), Schedule Report (Automated Daily/Weekly email delivery).

8. Module 7: Referral Program & Profit-Sharing Engine (Page 18)

7.1 Functional Specifications:

- **Refer & Earn Banner:** Prominent promo banner detailing: 'Earn 5% of your friend's profit per trade for up to 2 years.'
- **Referral Code & Share Widget:** Unique code card (e.g. `ST1234`), 1-click copy button, and 'Share & Invite' button (WhatsApp, SMS, Social apps).
- **Referral Earnings Overview Metrics:**
 - **Total Referral Earnings:** █ 12,450
 - **Available to Withdraw:** █ 8,750
 - **Pending (Settlement in Progress):** █ 3,700
 - **Total Referrals Count:** 8 active traders
- **Referrals Data Table (Page 18):**

Columns: Name, Active Status Badge, Joined Date, Total Trades Executed, Friend's Total Profit, Referrer Earnings (5%).
Example rows: Anil R (Active, █4,320 earned), Priya K (Active, █2,850 earned), Rahul S (Active, █1,780 earned), Meera T (Active, █1,250 earned).
- **Payout Action Buttons:**
 1. 'Withdraw Referral Earnings': Direct payout to registered bank account.
 2. 'Add to Master Account': Transfer referral balance into primary trading wallet balance.

9. Module 8: Funds Management & Withdrawal Workflow (Page 23)

8.1 Wallet Balances & Withdrawal Process:

- **Balance Display Strip:** Available Balance (█ 1,24,850), Withdrawable Amount (█ 1,20,000), Amount in Trades (█ 4,850).
- **Step 1 - Enter Withdrawal Amount:** Text input with quick-select chips (█ 5,000, █ 10,000, █ 25,000, █ 50,000, █ 1,000,000).

- **Step 2 - Select Registered Bank Account:** Dropdown / Card list of verified bank accounts showing Bank Logo, Account Number (masked `\*\*\*\* 4567`), IFSC Code, and 'Primary' badge.
- **Step 3 - Security Authentication Modal:** Modal window summarizing withdrawal details (Amount, Bank, IFSC, Processing Time: 1–4 hours). Requires authorization via **6-Digit PIN**, **Fingerprint**, or **Face ID**.
- **Step 4 - Withdrawal Status Tracker:** Tabbed view (*All | Processing | Completed | Failed*) listing request history with Txn ID (e.g., `TRW12345678`), Bank name, status badge, and date.

10. Module 9: Research, News & Advisory Calls (Page 25)

9.1 Research Hub Features:

- **Tabs:** Overview | News | Trading Calls | Reports.
- **Market News Feed:** Breaking news articles with source badge (e.g. Economic Times, Moneycontrol, Reuters, CNBC), timestamp, thumbnail image, and **Market Impact Sentiment Badge** (*Market Impact: Positive [Green], Neutral [Blue], Negative [Red]*).
- **Expert Advisory Trading Calls:** Signal cards providing actionable recommendations:
 - **NIFTY 50:** BUY recommendation | Target 1: 25,250 | Target 2: 25,350 | Stop Loss: 24,950 | Risk: Moderate | Type: Intraday
 - **RELIANCE:** BUY recommendation | Target 1: 2,950 | Target 2: 2,980 | Stop Loss: 2,870 | Risk: Moderate | Type: Swing
 - **BANK NIFTY:** SELL recommendation | Target 1: 51,200 | Target 2: 50,900 | Stop Loss: 51,750 | Risk: High | Type: Intraday
 - **GOLD OCT FUT:** SELL recommendation | Target 1: 72,400 | Target 2: 72,200 | Stop Loss: 73,150 | Risk: Moderate | Type: Intraday

11. Module 10 & 11: Help Support & Notifications System (Pages 21, 22)

10.1 Help & Support Center (Page 21):

- **Search Input:** Search help topics ('Search for help e.g. withdrawal, KYC, algo trading...').
- **Category Grid:** Account, KYC Verification, Algo Trading Strategies, Manual Trading Orders, Funds & Margin, Withdrawals, Referrals, Security & 2FA.
- **Support Channels:** **24x7 Live Chat Support** (Instant modal), **Call Support** (Toll-free 1800 123 4567), **Raise a Ticket** (Form modal with response within 4 hours).
- **FAQ Accordion:** Collapsible answers for common questions.

11.1 Notifications Center (Page 22):

- **Category Filters:** Trading (5 unread), Funds (2 unread), Account (1 unread), Important (3 unread).
- **Notification Cards:** Grouped by Date (Today, Yesterday, 10 Sep 2026). Real-time events for Order Execution, Algo Execution, Position Closed, Algo Daily Summary, and Deposit/Withdrawal updates.

12. Technical Architecture, Database Models & API Specifications

12.1 PostgreSQL Database Schema (Sequelize Models):

Sequelize Model	Attributes / Column Schema	Relationships & Constraints
User	id (UUID PK), full_name, email, mobile_number, password_hash, pin_hash (6-digit), pattern_hash, biometric_credentials, is_kyc_verified, referral_code, referrer_id	HasOne OnboardingDetails, HasMany BankAccounts, HasMany Orders, HasMany Transactions
OnboardingDetails	id, user_id (FK), dob, gender, pan_number, aadhaar_number, ky_document_url, address, city, state, pincode, occupation, income_range, experience, risk_profile, status	BelongsTo User
BankAccount	id, user_id (FK), account_holder_name, account_number, ifsc_code, bank_name, is_primary	BelongsTo User, HasMany Transactions
UserTradingSetup	id, user_id (FK), selected_package (Capital Protect/Balanced/Aggressive), selected_markets (JSON), algo_enabled (Boolean)	BelongsTo User
Position / Trade	id, user_id (FK), symbol, exchange, order_type, side (BUY/SELL), quantity, buy_price, ltp, pnl_amount, pnl_percent, is_algo (Boolean), status	BelongsTo User
Transaction	id, user_id (FK), bank_account_id (FK), type (WITHDRAWAL/DEPOSIT/REFERRAL_PAYOUT), amount, status (PROCESSING/COMPLETED/FAILED), reference_id	BelongsTo User, BelongsTo BankAccount
ReferralEarning	id, referrer_id (FK), referee_id (FK), trade_id, profit_amount, earned_amount (5%), status	BelongsTo User (Referrer & Referee)

12.2 RESTful API Endpoints Mapping:

- **Authentication:** `POST /api/auth/register`, `POST /api/auth/login`, `POST /api/auth/pin-login`, `POST /api/auth/biometric-login`, `GET /api/user/security-history`
- **Onboarding:** `POST /api/onboarding/step-1` to `step-7`, `POST /api/onboarding/upload-kyc`, `GET /api/onboarding/status`
- **Dashboard & Algo:** `GET /api/dashboard/summary`, `POST /api/algo/toggle`, `GET /api/algo/live-positions`, `GET /api/algo/performance`
- **Markets & Watchlist:** `GET /api/markets/watchlist`, `GET /api/markets/search?q=...`, `GET /api/markets/ticker`
- **P&L; & Reports:** `GET /api/pnl/summary`, `GET /api/pnl/positions`, `GET /api/pnl/export?format=pdf`
- **Referrals:** `GET /api/referrals/overview`, `POST /api/referrals/withdraw`
- **Funds & Withdrawals:** `GET /api/funds/balance`, `POST /api/funds/withdraw`, `GET /api/funds/history`
- **Research & News:** `GET /api/research/news`, `GET /api/research/calls`
- **Support & Notifications:** `GET /api/notifications`, `POST /api/support/ticket`

End of Requirements Specification Document — Travancore Research & Investments Ltd.